

EMBEDDED SYSTEMS LAB SETUP AND SMART IoT AUTOMATION SYSTEM

TASK-3

OBJECTIVE

The objective of this task is to upgrade the previous temperature monitoring system into a Smart IoT Automation System by implementing automation logic using an Arduino UNO and DHT11 sensor. The system automatically controls an LED based on the measured temperature and displays live readings through the Serial Monitor.

WORKING ENVIRONMENT

Environment Chosen: Wokwi Online Simulator

Microcontroller: Arduino UNO

Sensor: DHT11 Temperature & Humidity Sensor

Output Device: LED

Programming Language: Embedded C/C++ (Arduino)

COMPONENTS USED

- Arduino UNO
- DHT11 Temperature Sensor
- LED
- 220Ω Resistor
- Breadboard (Virtual)
- Jumper Wires
- Wokwi Simulator

CIRCUIT CONNECTIONS

	<u>DHT11 Pin</u>	<u>Arduino Pin</u>
VCC		5V
GND		GND
DATA		Digital Pin 2

LED Connections :

- **Positive (+)** → Digital Pin 13 through 220Ω resistor
- **Negative (-)** → GND

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WORKING PRINCIPLE

The DHT11 sensor continuously measures the surrounding temperature.

The Arduino reads the sensor data and compares it with a predefined threshold value.

- If the temperature is **greater than 30°C**, the LED turns **ON** indicating a high-temperature condition.
- Otherwise, the LED remains **OFF**.

The temperature values and system status are displayed on the Serial Monitor in real time.

AUTOMATION LOGIC

```
IF Temperature > 30°C
  LED ON
  Display "High Temperature Alert"
ELSE
  LED OFF
  Display "Temperature Normal"
```

ARDUINO PROGRAM

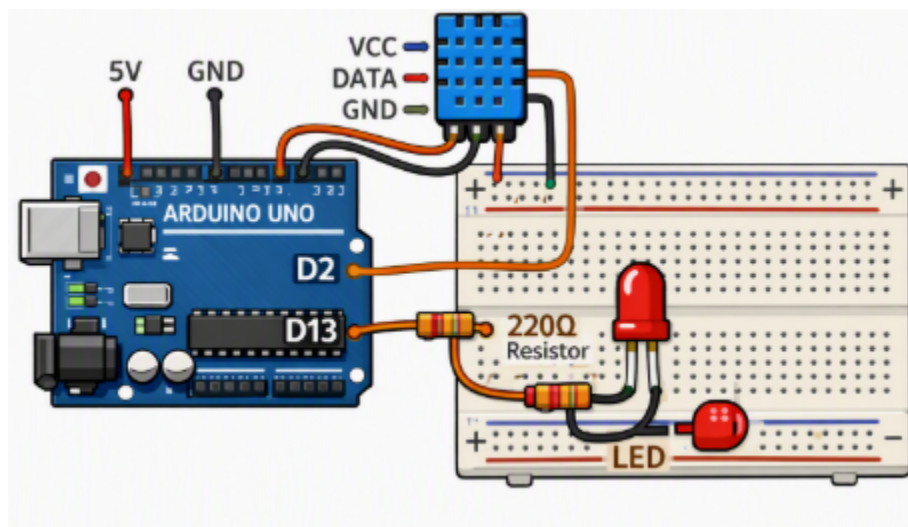
```
1  #include <DHT.h>
2
3  #define DHTPIN 2
4  #define DHTTYPE DHT11
5
6  DHT dht(DHTPIN, DHTTYPE);
7
8  const int ledPin = 13;
9
10 void setup() {
11     Serial.begin(9600);
12     dht.begin();
13     pinMode(ledPin, OUTPUT);
14 }
15
16 void loop() {
17
18     float temp = dht.readTemperature();
19
20     if (isnan(temp)) {
21         Serial.println("Sensor Error");
22         delay(2000);
23         return;
24     }
```

```
25
26     Serial.print("Temperature: ");
27     Serial.print(temp);
28     Serial.println(" °C");
29
30     if (temp > 30) {
31         digitalWrite(ledPin, HIGH);
32         Serial.println("Status : HIGH TEMPERATURE");
33         Serial.println("LED : ON");
34     } else {
35         digitalWrite(ledPin, LOW);
36         Serial.println("Status : NORMAL");
37         Serial.println("LED : OFF");
38     }
39
40     Serial.println("-----");
41
42     delay(2000);
43 }
```

OUTPUT

```
Temperature : 27°C  
Status : NORMAL  
LED : OFF  
  
-----  
  
Temperature : 32°C  
Status : HIGH TEMPERATURE  
LED : ON
```

CIRCUIT DIAGRAM



OBSERVATIONS

- The DHT11 sensor successfully measured the ambient temperature.
- The Arduino continuously monitored the sensor readings.
- The automation logic correctly controlled the LED.
- Live temperature values were displayed on the Serial Monitor.
- The system automatically responded whenever the threshold value was exceeded.

APPLICATION

- Smart Home Temperature Control
- Server Room Monitoring
- Industrial Temperature Monitoring
- Greenhouse Automation
- IoT Environmental Monitoring
- Smart Office Automation

ADVANTAGES

- Smart Home Temperature Control
- Server Room Monitoring
- Industrial Temperature Monitoring
- Greenhouse Automation
- IoT Environmental Monitoring
- Smart Office Automation

RESULT

The Smart Temperature Control System was successfully implemented using an Arduino UNO and DHT11 sensor. The system continuously monitored the ambient temperature, displayed live readings on the Serial Monitor, and automatically activated an LED whenever the temperature exceeded **30°C**. The project demonstrated the successful implementation of sensor interfacing, automation logic, and real-time monitoring.